

Taylor qatori

Reja:

1) Funksiyaning taylor qatoriga yoyish

2) Elementar funktsiyani taylor qatoriga yoyish

Biz yuqorida, har qanday darajali

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n + \dots$$

qator o'zining yaqinlashish intervali $(-r, r)$ da uzluksiz $S(x)$ funktsiyani (darajali qator yig'indisini) ifodalab, bu funktsiya shu oraliqda istalgan tartibdagi hosilaga ega bo'lishini ko'rdik. Endi biror oraliqda istalgan tartibdagi hosilaga ega bo'lgan funktsiyani darajali qatorga yoyish masalasini qaraymiz.

. Funktsiyalarni Taylor qatoriga yoyish.

$f(x)$ funktsiya $x = x_0$ nuqtaning biror

$$U_{\delta}(x_0) = \{x \in R: x_0 - \delta < x < x_0 + \delta\} \quad (\delta > 0)$$

atrofida berilgan bo'lib, shu atrofda funktsiya istalgan tartibdagi hosilaga ega bo'lsin.

Taylor formulasi

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + r_n(x)$$

ni yozish mumkin, bunda $r_n(x)$ qoldiq had. Berilgan $f(x)$ funktsiyaning x_0 nuqtada istalgan tartibdagi hosilaga ega bo'lishi Taylor formulasidagi hadlarning sonini har qancha katta olish imkonini beradi. Binobarin, tabiiy ravishda ushbu

$$f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \dots$$

qator yuzaga keladi. Bu maxsus darajali qator bo'lib, uning koeffitsientlari $f(x)$ funktsiya va uning hosilalarining x_0 nuqtadagi qiymatlari orqali ifodalanadi. Odatda darajali qator $f(x)$ funktsiyaning Taylor qatori deb ataladi. Xususan, $x_0 = 0$ da quyidagicha bo'ladi

$$f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

Tabiiy ravishda quyidagi savol tug'iladi: qachon biror $U_\delta(0)$ oraliqda berilgan, istalgan tartibdagi hosilaga ega bo'lgan $f(x)$ funksiyaning Teylor qatori

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

shu oraliqda xuddi shu $f(x)$ ga yaqinlashadi.

Teorema. $f(x)$ funksiya biror $(-r, r)$ ($r > 0$) oraliqda istalgan tartibdagi hosilaga ega bo'lib, uning $x = 0$ nuqtadagi Teylor qatori

$$f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

bo'lsin. Bu qator $(-r, r)$ oraliqda $f(x)$ ga yaqinlashishi uchun $f(x)$ funksiya Teylor formulasi

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + r_n(x)$$

ning qoldiq hadi barcha $x \in (-r, r)$ da nolga intilishi $\left(\lim_{n \rightarrow \infty} r_n(x) = 0\right)$ zarur va etarli.

Zarurligi. Avvalo (14.37) qatorning koeffisientlari bilan (14.38) Teylor formulasidagi koeffisientlarning bir xil ekanligini ta'kidlaymiz. (14.37) qator yaqinlashuvchi bo'lib, uning yig'indisi $f(x)$ ga teng bo'lsin. U holda bu qatorning qisman yig'indisi

$$S_n(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n$$

uchun

$$\lim_{n \rightarrow \infty} S_n(x) = f(x) \quad (\forall x \in (-r, r))$$

bo'ladi. Undan esa

$$\lim_{n \rightarrow \infty} [f(x) - S_n(x)] = \lim_{n \rightarrow \infty} r_n(x) = 0$$

bo'lishi kelib chiqadi.

Etarliliigi. ($\forall x \in (-r, r)$) $\lim_{n \rightarrow \infty} r_n(x) = 0$ da bo'lsin. U holda quyidagicha

$\lim_{n \rightarrow \infty} [f(x) - S_n(x)] = 0$ bo'lib, undan $\lim_{n \rightarrow \infty} S_n(x) = f(x)$ esa bo'lishi kelib chiqadi. Bu esa (14.37) qator $(-r, r)$ da yaqinlashuvchi bo'lib, uning yig'indisi $f(x)$ ga teng bo'lishini, ya'ni

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

ekanligini bildiradi.

Odatda keyingi munosabat o'rinli bo'lsa, $f(x)$ funksiya Teylor qatoriga yoyilgan deb ataladi.

23-teorema. Agar $f(x)$ funksiya $(-r, r)$ oraliqda ($r > 0$) darajali qatorga yoyilgan bo'lsa:

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n + \dots$$

bu qator $f(x)$ funksiyaning Teylor qatori bo'ladi.

Isbot. 20-teorema va uning natijasiga ko'ra (14.39) darajali qator $(-r, r)$ oraliqda istalgan marta (hadma-had) differensiallanuvchi bo'lib,

$$f'(x) = 1 \cdot a_1 + 2 \cdot a_2x + 3 \cdot a_3x^2 + \dots + na_nx^{n-1} + \dots$$

$$f''(x) = 1 \cdot 2 \cdot a_2 + 2 \cdot 3 \cdot a_3x + \dots + n(n-1)a_nx^{n-2} + \dots$$

$$f'''(x) = 1 \cdot 2 \cdot 3 \cdot a_3 + \dots + n(n-1)(n-2)a_nx^{n-3} + \dots$$

.....

$$f^{(n)}(x) = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1)na_n + \dots$$

.....

bo'ladi. Keyingi tengliklarda $x = 0$ deb quyidagilarni topamiz

$$a_0 = f(0), \quad a_1 = \frac{f'(0)}{1!}, \quad a_2 = \frac{f''(0)}{2!}, \quad a_3 = \frac{f'''(0)}{3!}, \dots, \dots,$$

Natijada (14.39) qatorning ko'rinishi quyidagicha bo'ladi:

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

Quyida funksiyaning Teylor qatoriga yoyilishining etarli shartini ifodalovchi teoremani keltiramiz

24-teorema. $f(x)$ funksiya biror $(-r, r)$ oraliqda istalgan tartibdagi hosilaga ega bo'lsin. Agar shunday $M > 0$ soni mavjud bo'lsaki, barcha $\forall x \in (-r, r)$ hamda barcha $n = 0, 1, 2, \dots$ uchun

$$|f^{(n)}(x)| \leq M \quad (f_x^0 = f(x))$$

tengsizlik bajarilsa, u holda $(-r, r)$ oraliqda $f(x)$ funksiya Teylor qatoriga yoyiladi, ya'ni

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \dots + \frac{f^{(n)}(0)}{n!} x^n + \dots$$

Bo'ladi.

Adabiyotlar

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